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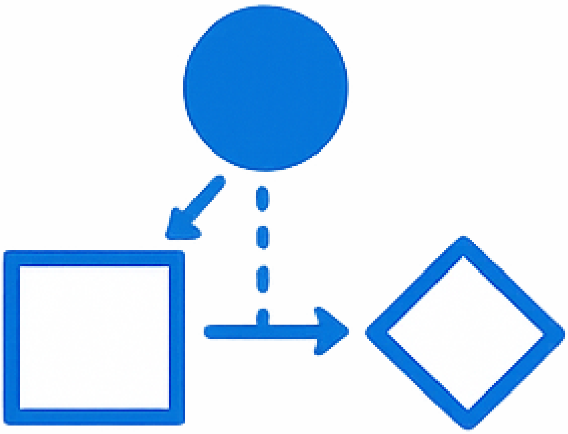

Assignment 2

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Qualitative Methods and Data Quality

Part I: Say-Do Gap Study and Part II: Radiology Data Quality Analysis

Date: March 2, 2026

Course: NDAB20010U Health Data and Interoperability

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AI Declaration

I declare that AI has only been used for grammatical correction.
AI has **not** been used to generate content, reasoning, or structure.

Contents

Abbreviation Key

Table 1: Internal abbreviations and identifiers used in this report

Abbreviation	Meaning in this report
Axx	Activity identifier in the interview-based model in Part I (for example, A01–A09).
Exx	Event identifier in the observed Part I event log (for example, E03, E10).
Dxx	Delta identifier in the Part I comparison sheet (for example, D01–D06).
Mx	Method identifier in the Part II data quality toolkit (for example, M2, M10).
RQ1	Research Question 1: workflow efficiency and bottlenecks.
RQ2	Research Question 2: timeliness of critical finding communication.
RQ3	Research Question 3: AI-based triage system development.
DQ	Data quality.
TAT	Turnaround time between selected workflow milestones.

1 Part I: A Real-World Study of the Say-Do Gap in Modelling

This part analyses one weekday morning routine across *work-as-imagined*, *work-as-done*, and *work-as-recorded*, with explicit links between claims and evidence.

1.1 Deliverable 1: Description of the Routine Based on the Interview

Scope and evidence This Part I study is a single-case self-study with one participant (the author) and one routine (weekday morning preparation before leaving home). The interview was conducted on February 18, 2026 as a semi-structured self-interview.

Following Chapter 27, this step is treated as *manual knowledge acquisition*: it captures *knowledge-as-stated* (what the participant reports) and produces an initial *work-as-imagined* model (how the routine is expected to unfold) (Guide to Health Informatics 2026; Zajac 2026). Internal identifiers used in the report are summarised in Table 1.

The routine boundary is anchored in transcript evidence (Appendix A, Table 3, lines L21–L22):

- Start: the alarm sounds.
- End: the apartment door closes when leaving.

The baseline sequence is grounded in lines L02–L12: wake-up, hygiene, dressing/packing, breakfast, final checks, and departure. Variation points (snooze, breakfast substitution, key search, occasional roommate coordination) are grounded in lines L14–L18.

Participants, order, and modelling assumptions Table 4 presents the interview-based activity model, and Figure 1 shows the corresponding interview-based DCR graph. In this context, interview-based DCR means a process model built from interview evidence rather than direct observation.

Assumptions used in Deliverable 1 are:

- The model represents a weekday baseline, not weekend behaviour.
- The participant is the only consistent human actor in the routine.
- Expected times are recall-based estimates and therefore approximate.

Together, these assumptions keep scope explicit and make later comparison with observed evidence in Deliverables 2 and 3 auditable.

1.2 Deliverable 2: Event Log Representing the Observed Routine

The observation was conducted on February 20, 2026 (06:45–07:30), two days after the interview. This delay was intentional to reduce recall carry-over between elicitation and observation.

In methodological terms, the observation captures *work-as-done* (what was actually performed in context). The resulting CSV captures *work-as-recorded* (what is represented in structured trace data) (Guide to Health Informatics 2026; Zajac 2026). To keep data collection transparent, the observation followed a short guide (Appendix A, Table 2).

The submitted event log includes the fields `case_id`, `event_id`, `timestamp`, `event_label`, `actor`, `tool`, `context_note`, and `related_activity_id`. The full machine-readable file is submitted separately, while Appendix A, Table 5 shows the key excerpt used in the report.

Compared with the interview baseline, the observation reveals three anchor differences that are used in Deliverable 3:

- E03: extra coordination event,
- E09: order shift relative to the interview sequence,
- E10: explicit workaround behaviour.

These observations provide the empirical basis for the delta analysis between stated and observed practice.

1.3 Deliverable 3: Delta Sheet Summarising Differences Between Stated and Observed Practice

The comparison maps interview activities (Axx identifiers) to observed events (Exx identifiers). A delta is registered when at least one of the following differs: event presence, order, timing, workaround behaviour, or activity interpretation. This follows the Chapter 27 logic of comparing observed traces with a prior process description to expose model limits and recurring exceptions (Guide to Health Informatics 2026).

Appendix A, Table 6 contains the complete delta sheet. The six required categories are all represented:

- Missing event: no distinct seated breakfast-and-schedule-review step.
- Extra event: roommate coordination appears only in observed data.
- Different order: bag packing occurs after breakfast preparation starts.
- Timing distortion: hygiene begins later than the interview baseline.
- Workaround/coordination step: key/lock issue triggers ad hoc messaging.

- Relabeling/ambiguity: *breakfast* changes from sit-down meal to banana plus coffee-to-go.

These deltas operationalise the say-do gap, meaning the difference between described routine and performed routine. They also prepare the reflection in Deliverable 4 by showing where model assumptions break under real execution conditions.

1.4 Deliverable 4: Reflections on the Modelling Process

1. **What kinds of data were missed or misrepresented in the interview-based model, and why?**

   The interview-based model captured the participant's general understanding of the routine, but it missed several details that only became visible during observation. In particular, it underrepresented small coordination steps, local adjustments, and recovery behaviour when something did not go as planned. For example, the observed routine included micro-coordination with a roommate (E03), a workaround related to keys and locking (E10), and no clearly separated seated breakfast-and-schedule step, even though this appeared in the interview-based account. This happened because interviews mainly produce *knowledge-as-stated*: they rely on memory, simplification, and what seems important enough to mention, rather than on what is actually done moment by moment in context.

2. **What counts as an activity in each model, and how did this differ?**

   In the interview-based model, an activity is defined at the level of intention or routine description. The activities in Deliverable 1 therefore describe meaningful chunks such as getting ready, preparing breakfast, or doing final checks. In the observed model, by contrast, an activity is represented through timestamped events that can be directly seen and recorded. This produces a finer level of granularity, where one interview activity may correspond to several observed events. As a result, the interview-based DCR graph presents a more orderly and compressed routine, while the observation-based event log makes variation, overlap, and short coordination steps much more visible.

3. **How did the event log constrain what could be modelled?**

   The event log made it possible to model sequence, timing, and visible deviations with reasonable precision, but it also constrained the analysis to what could be explicitly recorded as events. It shows *what* happened and *when*, but it is weaker at showing motives, assumptions, hesitations, and tacit judgement. Some context could be added in notes, but this still does not fully explain why a deviation occurred or how the participant interpreted the situation in the moment. The event log therefore supports conformance-style comparison well, but it cannot by itself provide a complete account of the routine as lived practice.

4. **Do you think that the event log could be captured automatically by a system? Why or why not?**

   Parts of the event log could probably be captured automatically, but not the full routine. System-based logging works well for digital traces such as phone use, timestamps, or app interactions. However, many relevant parts of this case were social, practical, or embodied rather than digital. Coordination with another person, searching for objects, or making a quick workaround decision would be difficult to capture reliably unless the environment were heavily instrumented. Even then, the system might still miss the meaning of the action. For that reason, automatic capture could support this kind of modelling, but it would not remove the need for qualitative observation.

5. **How did the observation help uncover variations or exceptions that were not mentioned in the interview?**

   The observation made it possible to see how the routine changed under actual conditions rather than idealised recall. Several differences appeared that were either absent from the interview or described more neatly than they happened in practice. These included changes in order, the absence of a distinct A06 step, and the workaround captured in E10. Observation was useful because it exposed exceptions as part of normal performance rather than as rare anomalies. In this sense, it showed that the gap between stated and observed practice was not just about memory error, but also about how routines are continually adjusted in context.

6. **Why is it relevant to consider qualitative methods of discovery when building process models?**

   Qualitative methods such as observation are relevant because process models are not neutral mirrors of reality; they are constructed representations shaped by the data used to build them. If modelling relies only on interviews or logs, important aspects of practice may disappear, especially informal coordination, situated judgement, and practical adaptations. In this study, combining interview data, observation, and the event log produced a more valid account than any one source alone. This supports the Chapter 27 argument that model construction depends on how knowledge is discovered, and that robust process models benefit from triangulation across stated, observed, and recorded evidence (Guide to Health Informatics 2026; Zajac 2026).

2 Part II: Radiology Case Study — Data Quality, Conformance, and the Limits of Logs

This part briefly documents the separately submitted notebook artefact and presents the Part II reflection section required in the main report.

2.1 Notebook Artefact Submitted Separately

The completed notebook (`assignment.ipynb`) is submitted as a separate artefact and contains all required Part II notebook work (Parts 0–7), including method selection, executed analyses, the custom DQ dimension, and the invisible-work comparison.

Method identifiers (Mx) and research-question identifiers (RQ1–RQ3) follow Table 1.

2.2 Deliverable 6: Part II Reflections

1. **Initial exploration Q1.a Data type in the log**

   The file is a case-based radiology process event log. Each row is one event with case identifier, timestamp, activity label, actor, role, and optional clinical/technical context (e.g., modality, report-linked fields, notes). From exploration outputs in Appendix B, Table 7, the dataset contains 150 events, 17 cases, and 12 unique logged activities (activity list in Appendix B.5, Table 12).

   Q1.b RQ use under good data quality

   If quality is good, the same structure supports all three research questions:

   - **RQ1:** timestamps and activity order allow bottleneck and turnaround-time analysis.
   - **RQ2:** critical communication events can be checked for conformance and timeliness.
   - **RQ3:** event traces can be transformed into candidate features for triage-model development.

   The log is therefore fit-for-purpose in principle, but only under acceptable completeness, representativity, timeliness, and provenance quality.

2. **Method selection justification Q2.a Completed method-selection table**

   The completed table is included in Appendix B, Table 8. Selected methods: **M2, M6, M8, M10, M11**.

   Q2.b Selected methods and rationale

   The selected set was chosen to cover all RQs with minimal overlap:

- **M2:** baseline completeness gate before downstream interpretation.
- **M6:** representativity of urgency/shift context for generalisability.
- **M8:** direct safety-critical communication test (RQ2 core).
- **M10:** performance metrics with timestamp-credibility sensitivity (RQ1 core).
- **M11:** workload concentration patterns relevant to RQ1 and RQ3.

Q2.c Deprioritised methods and rationale

Methods were deprioritised mainly because of scope and overlap:

- **M1:** useful schema check, but lower analytical value after basic integrity is confirmed.
- **M3:** ICD-11 format checks are useful, but missingness reduced immediate explanatory gain.
- **M4:** partly overlaps M11; M11 gave clearer workload concentration outputs.
- **M5:** modality case-mix useful, but less central than urgency/shift representativity for this report.
- **M7:** temporal-conformance value partly overlapped M10 timestamp-credibility analysis.
- **M9:** clinically relevant, but less central for cross-RQ synthesis in the selected scope.

Q2.d Trade-offs considered

Main trade-off: breadth versus depth. Running five methods gave full RQ coverage while keeping interpretation coherent rather than fragmented across too many partially redundant outputs.

Q2.e RQ1–RQ3 influence on selection

RQ1 drove M10 and M11, RQ2 drove M8, and RQ3 drove representativity/provenance emphasis (M6 plus interpretation from M2 and custom method). This ensured each RQ had at least one direct method and supporting cross-checks.

3. **Results interpretation** Key outputs are summarised in Appendix B, Table 9, with figures in Figures 2–5. Interpretation was mapped to the five-facet assessment lens (Data, Source, System, Task, Human) to keep method results comparable across RQs (Mohammed et al. 2025).

   Method 2 (M2: Completeness). High missingness in context fields (`study_uid` 134/150, `modality` 101/150) confirms that sequence-level analysis is feasible but clinically richer interpretation is constrained. This primarily informs Data, Source, and System facets; improvement requires stricter source-level field population rules.

Method 6 (M6: Representativity). Urgency and shift skew (15 stat/urgent vs 2 routine; Day 11, Night 6, Evening 0) shows limited representativity across operational contexts. Direction matched expectations, but the magnitude was larger than expected. Main facets: Data, Task, Human. Improvement: stratified sampling and/or longer capture windows.

Method 8 (M8: Critical communication conformance). Result: 16 pass and 1 fail (`CXR-004` missing critical communication event). Even with high aggregate pass rate, the failed case is a clinically significant safety non-conformance and should be treated as a high-priority process signal, not statistical noise. Main facets: Task, System, Human. Improvement: mandatory traceability for critical communication steps.

Method 10 (M10: TAT + batch-logging detection). Mean total TAT changed from 147.93 min (raw) to 198.50 min (clean), with 6/17 cases flagged (35.29%). This confirms that performance conclusions are highly sensitive to timestamp credibility and logging behaviour. Main facets: Data, System, Task. Improvement: explicit entry-mode metadata and stricter real-time logging controls.

Method 11 (M11: Workload distribution). Events were concentrated by role (radiographer 64, system 31, senior 31, clinician 17, resident 7), confirming uneven visibility and workload distribution. Main facets: Human, Task, System. Improvement: compare observed concentrations with planned staffing and shift allocations.

Across methods, the biggest cross-cutting insight is that conclusions become much less stable when representativity and timestamp credibility issues are accounted for.

4. **Proposed DQ dimension Q4.a Dimension definition**

The proposed custom DQ dimension is **operational authenticity**: whether logged traces represent real clinical workflow rather than test/training/administrative artifacts.

Q4.b Motivation from exploration evidence

In exploration code (Part 5), case `TEST-001` was flagged by multiple contamination signals: non-standard case ID, training/test marker in notes (`TRAINING CASE - DO NOT PROCESS`), administrative actor involvement, and a one-event trace. This pattern is not well covered by standard completeness/conformance checks but can bias downstream analysis.

Q4.c Custom DQ assessment code

The full implementation is provided in Appendix B.3, Listing 1.

Q4.d RQ impact and mechanism

Summary results (Appendix B.2, Table 10) show 1 high-risk flagged case out of 17 (5.88%). Even this small share affects all RQs:

- **RQ1:** can distort apparent workflow timing and throughput metrics.

- **RQ2:** can alter conformance denominators and bias pass/fail interpretation.
- **RQ3:** can contaminate model-development data and reduce validity.

Q4.e Facet-mapping table reference

Facet mapping is included in Appendix B.2, Table 11.

5. **Per-facet data quality synthesis Q5.note Custom-method code reference**

The custom assessment code used in this synthesis is documented in Appendix B.3, Listing 1.

Q5.a–e Facet findings, issues, and RQ consequences

Using the facet framing from Mohammed et al. (2025), the synthesis is:

- **Data facet:** M2 and M10 show high sparsity and timestamp-sensitive metrics; this weakens confidence in RQ1 performance estimates and RQ3 feature reliability.
- **Source facet:** custom-method flags (e.g., `TEST-001`) plus context sparsity indicate provenance contamination risk; this lowers transferability and trust, especially for RQ3.
- **System facet:** M8/M10 show that logging design and entry behaviour shape visible process reality; this can bias both RQ1 and RQ2 when logged time is treated as exact work time.
- **Task facet:** M6/M8 show urgency-shift imbalance and threshold dependence; this means efficiency and safety conclusions may partly reflect case mix and policy configuration.
- **Human facet:** M11 (plus M10/custom indicators) shows concentration and documentation-pattern effects; this can produce role-biased bottleneck and AI-readiness interpretations (RQ1, RQ3).

Q5.f Most challenging facet and reason

The **Human** facet was most difficult. Event logs capture actions but rarely intent, negotiation, and informal coordination. The **Source** facet was also difficult because provenance questions often require context outside the event table.

Q5.g Alternative assessment options

- Add explicit provenance metadata (production/test flag, entry mode, user context).
- Strengthen interoperability at capture time with a consistent event profile (stable case identifiers, structured reason codes, and aligned FHIR resource/event mappings such as `ServiceRequest`, `ImagingStudy`, `DiagnosticReport`) to reduce cross-system ambiguity (Benson and Grieve 2016; Hildebrandt 2026).

- Triangulate with linked systems (RIS/PACS/message logs) for cross-source validation.
- Use targeted mini-interviews or debriefs for anomalous cases.
- Run periodic audits on documentation practice and timing behavior.

6. **Invisible work and system design implications Q6.a Invisible-work table reference**

   The completed table is included in Appendix B.5, Table 13, grounded in workflow Figure 6 (Appendix B.4). Included activities: radiologist-to-radiologist consultation, clarification call to ordering clinician, informal reprioritisation/handoff, and cross-team coordination on patient readiness.

   Q6.b Why activities may be unlogged (technical/social/economic)

   Across the listed activities, missingness is plausibly explained by:

   - **Technical:** phone/verbal and cross-system interactions often lack a shared event schema.
   - **Social:** informal coordination is situational and intentionally lightweight.
   - **Economic/workload:** complete micro-documentation increases workload and may reduce throughput.

   Q6.c Log-only model limitations and RQ consequences

   If modeling relies only on logged events, the model misses real coordination pathways:

   - **RQ1:** bottlenecks can be misattributed because hidden coordination delays are not represented.
   - **RQ2:** safety communication may appear stronger than practice if verbal escalation is invisible.
   - **RQ3:** AI models may learn an oversimplified process missing key real-world coordination behavior.

   Q6.d Part I parallel: interview vs observation

   This mirrors Part I directly: interview captured work-as-imagined, while observation captured work-as-done with deviations and ad hoc coordination. Here, event logs capture work-as-recorded, which is again only a partial view of practice.

   Q6.e Implication for data-driven healthcare IT

   The gap shows that data-driven design is necessary but insufficient on its own. In Chapter 9 terms, formal systems encode codified work, but clinically important informal coordination can remain outside system boundaries (Guide to Health Informatics 2026; Guide to Health Informatics 2026).

Q6.f Concrete logging change, trade-offs, and mitigation

One concrete change is a lightweight `coordination_interaction` event type at high-impact points (e.g., escalation, reprioritisation, cross-team handoff), with minimal fields: timestamp, linked case ID, initiating role, receiving role, reason code. Trade-offs: more documentation burden and possible surveillance concerns. Mitigation: selective (not universal) capture, role-based visibility, and governance that uses logs for process improvement rather than individual blame.

A Supporting Tables for Part I

This appendix contains supplementary Part I material referenced from Deliverables 1–4.

Table 2: Observation guide used during Deliverable 2

Focus	Observation prompt	Evidence target
Sequence and boundary	Which event marks start/end, and what is the observed order of actions?	Event IDs, timestamps, sequence notes
Tools and artefacts	Which tools/resources are used at each step?	Tool field, context notes
Interactions and coordination	Does any interaction with other people occur?	Actor field, coordination notes
Deviations from interview baseline	Which actions are skipped, added, reordered, or relabeled relative to A01-A09?	Related activity ID + delta notes
Adaptation and recovery	Which workaround or exception handling occurs under time/resource pressure?	Context notes for exception events

Source: Zajac (2026) and Guide to Health Informatics (2026a): interview + observation + process-tracing principles adapted to this routine study.
Note: Field notes were split into descriptive and reflexive notes; only descriptive notes were transcribed into the submitted CSV event log.

A.1 Core Deliverable Artefacts (Part I)

Table 3: Line-numbered self-interview transcript (conducted February 18, 2026)

Line	Transcript
L01	Interview question: Please describe your normal weekday morning routine from when you wake up until you leave home.
L02	Participant: My alarm goes off at 06:45, and I usually get up right away after stopping it.
L03	Participant: I first drink a glass of water, then I go to the bathroom.
L04	Participant: I normally take a quick shower, around ten to twelve minutes.
L05	Participant: After that I get dressed and prepare my bag for the day.
L06	Participant: I usually pack my laptop, charger, student ID card, wallet, and keys.
L07	Participant: Then I make coffee and toast.
L08	Participant: I normally sit down, eat breakfast, and check my calendar and transport timing on my phone.
L09	Participant: After breakfast I brush my teeth and do a final mirror check.
L10	Participant: I check the weather so I can choose the right jacket.
L11	Participant: Before leaving, I do a final essentials check: phone, wallet, keys, and card.
L12	Participant: I aim to leave at around 07:30 to catch transport to campus.
L13	Interview question: Which steps vary the most from day to day?
L14	Participant: Sometimes I snooze once if I am tired.
L15	Participant: Breakfast can change; if I am in a rush, I might just eat fruit instead of toast.
L16	Participant: Occasionally I spend a minute looking for keys if I did not put them in the same place.
L17	Interview question: Are other people typically involved in this routine?
L18	Participant: Mostly no, I do it alone, but once in a while I text my roommate if I need a shared item.
L19	Interview question: Which tools or systems are central in the routine?
L20	Participant: My phone alarm, calendar, weather app, the coffee machine, toaster, and my backpack.
L21	Interview question: How do you define start and end of this routine?
L22	Participant: It starts when the alarm sounds and ends when I close the apartment door behind me.

Table 4: Interview-based activity model (work-as-imagined / knowledge-as-stated)

ID	Activity label	Actor	Order	Expected time	Main tools/resources
A01	Stop alarm and orient to time	Participant	1	06:45-06:47	Smartphone alarm
A02	Drink water	Participant	2	06:47-06:50	Glass, tap
A03	Toilet and shower	Participant	3	06:50-07:02	Bathroom, shower
A04	Dress and pack bag	Participant	4	07:02-07:12	Wardrobe, backpack
A05	Prepare breakfast and coffee	Participant	5	07:12-07:20	Coffee machine, toaster
A06	Eat breakfast and review schedule	Participant	6	07:20-07:25	Phone calendar/transport apps
A07	Brush teeth and final grooming	Participant	7	07:25-07:28	Toothbrush, mirror
A08	Check weather and essentials	Participant	8	07:28-07:30	Weather app, keys, wallet, ID
A09	Leave apartment for commute	Participant	9	07:30	Front door, bag

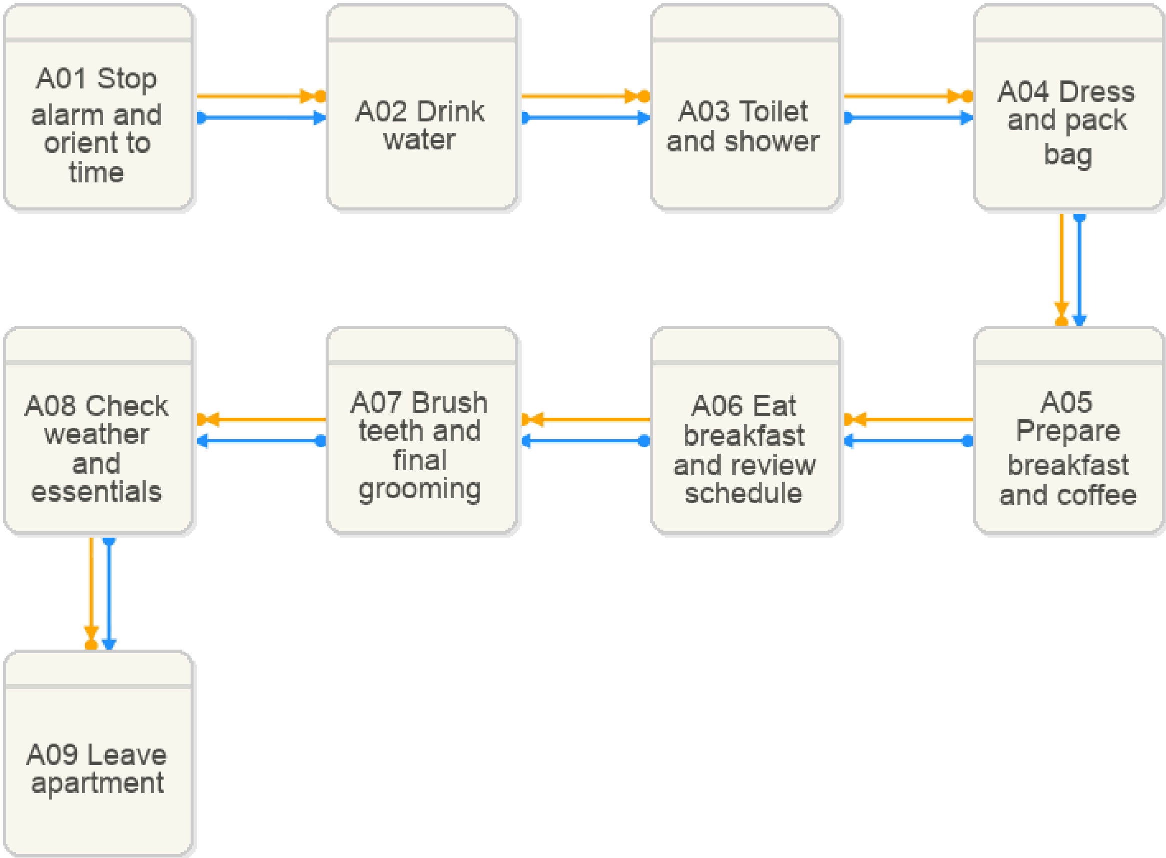

Figure 1: Interview-based DCR graph for the morning routine (A01-A09)

Source: Own DCR Graph export (February 2026), built from transcript evidence in Table 3 (lines L02-L12, L14-L18, L21-L22).

Note: Used as baseline work-as-imagined model for comparison with observed events in Deliverables 2 and 3.

Table 5: Observed event-log excerpt (February 20, 2026)

Event	Timestamp	Event label	Tool	Context note	Ref.
E01	2026-02-20 06:45	Alarm rings and snooze	Smartphone alarm	One snooze before wake-up.	A01
E03	2026-02-20 06:52	Reply to roommate message	Messaging app	Extra micro-coordination event.	NA
E05	2026-02-20 06:57	Toilet and shower	Bathroom	Hygiene block started later than interview expectation.	A03
E07	2026-02-20 07:12	Start coffee and toast	Coffee machine, toaster	Breakfast preparation begins.	A05
E08	2026-02-20 07:16	Switch breakfast to banana	Kitchen	Breakfast adapted due to no bread.	A05
E09	2026-02-20 07:18	Pack bag	Backpack	Packing occurs after breakfast prep starts.	A04
E10	2026-02-20 07:21	Find keys and borrow lock	Keys, messaging app	Workaround to solve missing item.	A08
E14	2026-02-20 07:30	Leave apartment	Front door	End of observed routine boundary.	A09

Table 6: Delta sheet between interview model and observed routine

Delta ID	Category	Interview ref.	Observed ref.	Difference description	Implication
D01	Missing events	A06	No direct event	No distinct seated breakfast-and-schedule-review event occurred.	Interview model overstates routine regularity.
D02	Extra events	None in baseline	E03 (06:52)	Short roommate-message coordination appears in observed routine only.	Reveals practical coordination load.
D03	Different order	A04 before A05	E07 then E09	Bag packing happened after breakfast preparation started.	Ordering constraints are weaker than stated.
D04	Timing distortions	A03 expected 06:50-07:02	E05 start 06:57	Hygiene block shifted later due to wake-up delays.	Time windows in interview are optimistic.
D05	Workarounds/coordination steps	A08 final check	E10 (07:21)	Missing lock triggered ad hoc borrowing via message.	Recovery actions are underrepresented in interview model.
D06	Relabeling/ambiguity	A05/A06	E08, E13	“Breakfast” changed from sit-down meal to banana plus coffee-to-go.	Label definitions need operational clarity.

B Supporting Tables and Code for Part II

This section includes only the relevant outputs from the completed Deliverable 5 notebook (`assignment.ipynb`): selected method evidence, key visualisations, custom DQ-dimension results, and invisible-work comparison material.

Table 7: Part II log profile from notebook Part 1 exploration

Metric	Result
Total events	150
Unique cases	17
Time range covered	2026-01-12 07:45 to 2026-01-13 08:46
Unique logged activities	12
Columns with strongest missingness	`study_uid` (134), `series_count` (134), `instance_count` (134), `icd11_code` (131), `report_id` (113), `modality` (101), `notes` (73)

Table 8: Selected methods run in notebook Part 4

Method	DQ dimension	Primary facets	RQ link	Selection rationale
M2	Completeness	Data, Source, System	RQ1–RQ3	Baseline quality gate before downstream interpretation.
M6	Representativity	Data, Task, Human	RQ1, RQ3	Checks urgency and temporal balance of case coverage.
M8	Timeliness conformance	Task, System, Human	RQ2	Tests safety-critical communication behavior directly.
M10	Timeliness + credibility	Data, System, Task	RQ1	Quantifies TAT and sensitivity to batch logging.
M11	Workload representativity	Human, Task, System	RQ1, RQ3	Reveals concentration across actors/roles.

Table 9: Key quantitative outputs from selected methods

Method	Key output	Relevance for Deliverable 6 reflection
M2	Context-field missingness is high (e.g., `study_uid` 134/150; `modality` 101/150).	Shows what can and cannot be trusted for workflow and AI-oriented interpretations.
M6	Urgency skew (15 stat/urgent, 2 routine) and shift imbalance (Day 11, Night 6, Evening 0).	Demonstrates representativity limits for RQ1 and especially RQ3 generalisation.
M8	Critical communication: 16 pass, 1 fail (`CXR-004` missing communication event).	Direct patient-safety signal tied to RQ2 and conformance assumptions.
M10	Mean total TAT = 147.93 min raw vs 198.50 min clean; 6/17 cases flagged as batch-logged (35.29%).	Shows strong sensitivity of performance conclusions (RQ1) to timestamp credibility.
M11	Events per role: radiographer 64, system 31, senior 31, clinician 17, resident 7.	Quantifies workload concentration and role imbalance relevant to RQ1 and RQ3.

B.1 Notebook Visualisations (Relevant Outputs)

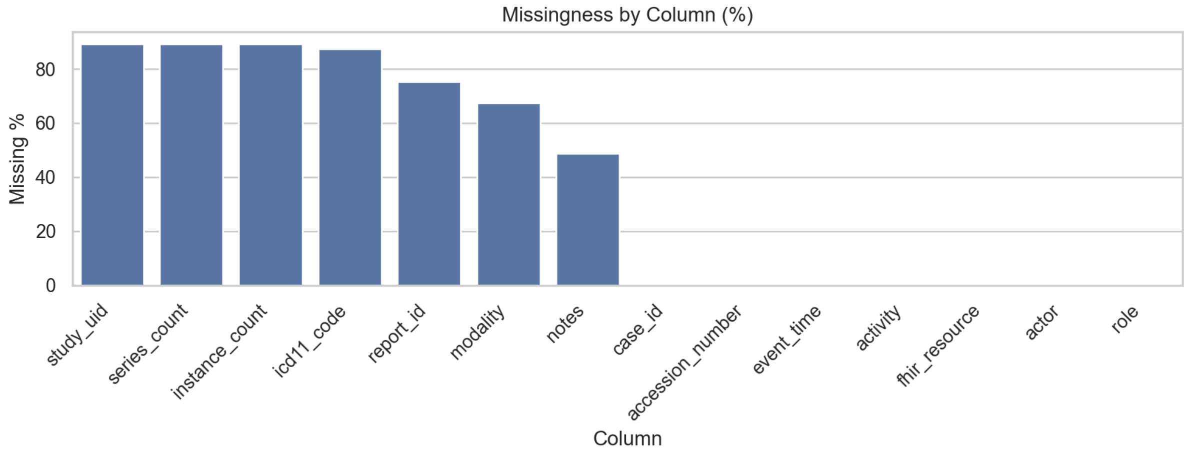

Figure 2: Method 2 output: missingness profile by column

Source: Generated from Deliverable 5 notebook output using `radiology_event_log.csv`*.*

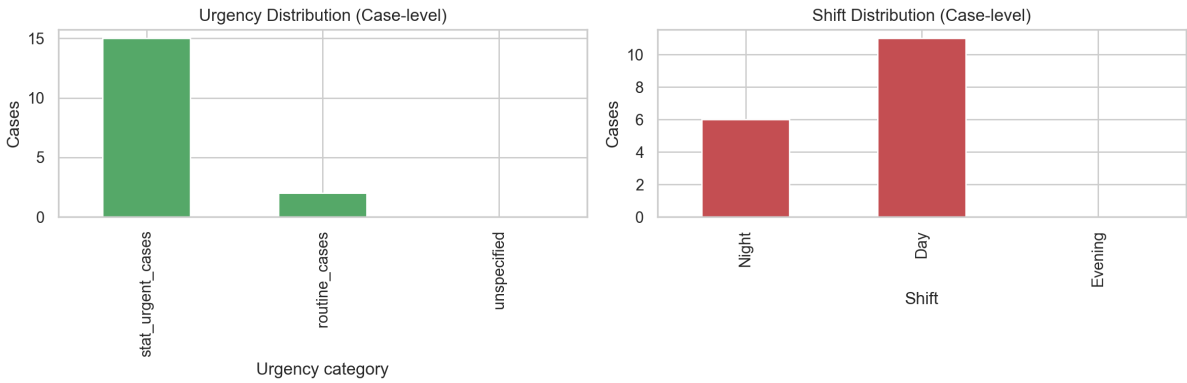

Figure 3: Method 6 output: urgency and shift representativity

Source: Generated from Deliverable 5 notebook output using `radiology_event_log.csv`*.*

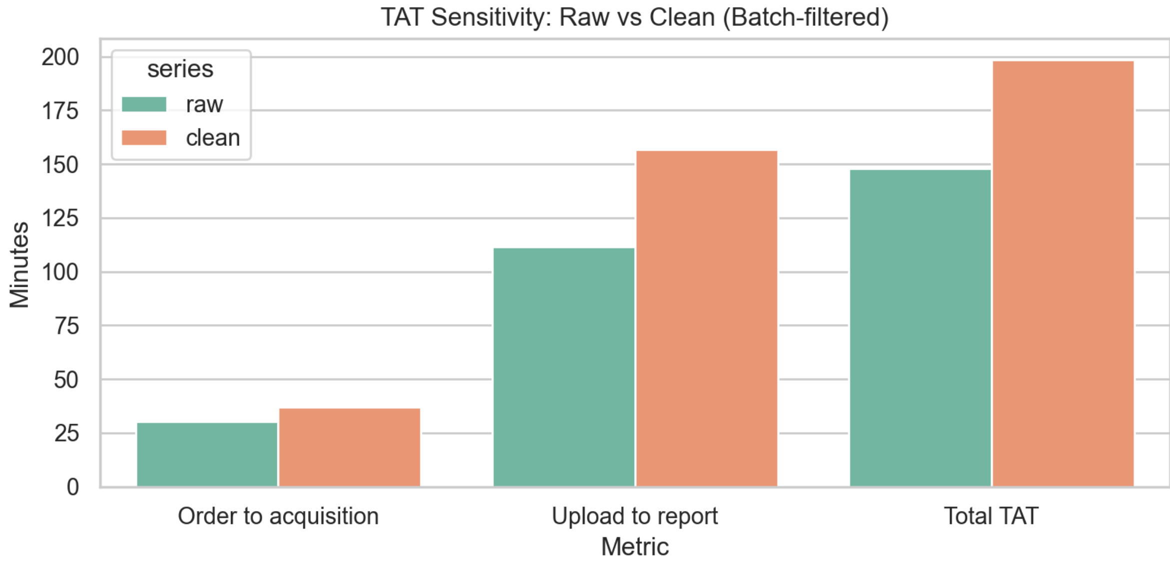

Figure 4: Method 10 output: turnaround-time sensitivity (raw vs batch-filtered)

Source: Generated from Deliverable 5 notebook output using `radiology_event_log.csv`.

Figure 5: Method 11 output: workload distribution by role

Source: Generated from Deliverable 5 notebook output using `radiology_event_log.csv`.

B.2 Custom DQ Dimension (Notebook Part 5)

Table 10: Custom DQ dimension result summary: operational authenticity

Metric	Result
Total evaluated cases	17
Flagged cases (medium/high risk)	1
Flagged case rate	5.88%
High-risk cases	1
Flagged case details	TEST-001: unexpected case ID format; training/test marker in notes; administrative actor involvement; extremely short trace (1 event)

Table 11: Facet mapping for custom dimension: operational authenticity

Facet	Involvement	Justification
Data	++	Case identifiers and note text contain direct contamination signals (test/training traces).
Source	++	Source governance determines whether non-production traces enter the analysis dataset.
System	+	Logging architecture does not fully separate test and production provenance.
Task	+	KPI and triage conclusions become biased when non-clinical traces are included.
Human	+	Human setup/maintenance actions can introduce non-operational events.

B.3 Custom DQ Assessment Method Code

Listing 1: Custom DQ assessment method from notebook Part 5 (operational authenticity)

```python
def my_custom_dq_method(df: pd.DataFrame) -> Dict[str, any]:
    """
    Assess operational authenticity (production-relevance validity).

    Goal:
    Identify case traces that are likely test/training/non-production records
    or otherwise not representative of real clinical workflow data.
    """
    rows = []

    for case_id, case_df in df.groupby('case_id'):
        reasons = []

        if not re.match(r'^CXR-\d{3}$', str(case_id)):
            reasons.append('Unexpected case_id format')
```

```

        if case_df['notes'].astype(str).str.contains(
            r'training|test|do not process', case=False, na=False
        ).any():
            reasons.append('Training/test markers in notes')

        if case_df['actor'].eq('Admin User').any():
            reasons.append('Administrative/test actor involvement')

        if len(case_df) < 3:
            reasons.append('Extremely short trace (low operational realism)')

        score = len(reasons)
        if score >= 3:
            risk_level = 'high'
        elif score >= 1:
            risk_level = 'medium'
        else:
            risk_level = 'low'

        rows.append({
            'case_id': case_id,
            'risk_level': risk_level,
            'risk_score': score,
            'reasons': '; '.join(reasons) if reasons else 'None',
            'event_count': len(case_df),
        })

    case_flags = pd.DataFrame(rows).sort_values(
        ['risk_score', 'case_id'], ascending=[False, True]
    )
    flagged = case_flags[case_flags['risk_level'].isin(['high', 'medium'])].copy()

    summary_metrics = {
        'total_cases': int(len(case_flags)),
        'flagged_cases': int(len(flagged)),
        'flagged_case_rate_pct': round(
            100 * len(flagged) / len(case_flags), 2
        ) if len(case_flags) > 0 else 0.0,
        'high_risk_cases': int((case_flags['risk_level'] == 'high').sum()),
    }

    return {
        'case_flags': case_flags,
        'flagged_cases': flagged,
        'summary_metrics': summary_metrics,
    }
```

B.4 Workflow Diagram from Assignment Brief

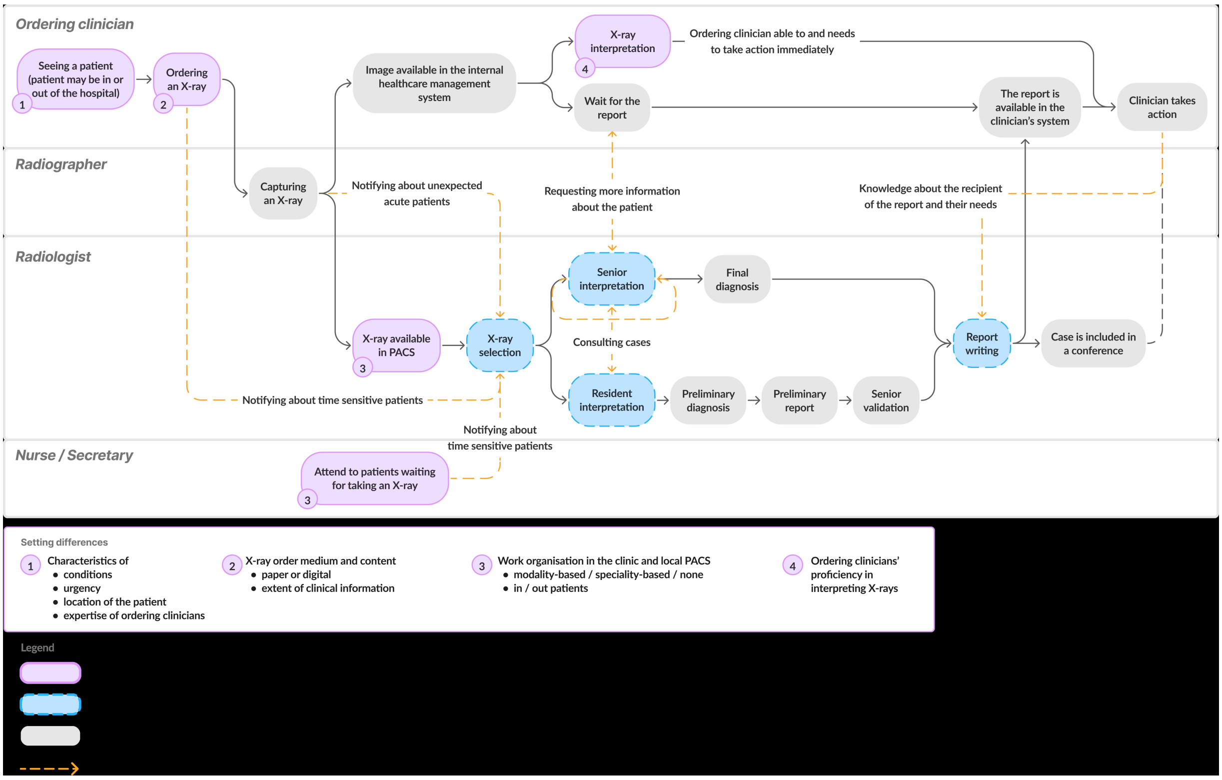

Figure 6: Assignment Figure 1 from the Assignment 2 PDF: observed radiology workflow with formal and informal coordination paths

Source: Provided in Assignment 2 brief as Figure 1 (`Figur1-assignement2.jpg`).

B.5 Invisible Work Material (Notebook Part 7)

Table 12: Activities logged in the event log (12 unique activities)

Logged activity set A	Logged activity set B
Case selected	Order placed
Critical communication	Patient selected from worklist
Image acquisition start	Preliminary report created
Image acquisition end	Report distributed
Images uploaded to PACS	Report finalised
Order accepted	Report validated and approved

Table 13: Invisible-work activities from workflow diagram not represented explicitly in log

Activity from diagram	Why not logged?	Facet	Consequence	Capture option
Radiologist-to-radiologist consultation	Mostly verbal/ad hoc interaction outside structured capture	Human, Task	Collaborative interpretation work is under-represented	Optional lightweight consult event
Clarification call to ordering clinician	Usually done by phone outside RIS/PACS event schema	System, Source	Communication-delay attribution becomes incomplete	Linked communication event type
Informal reprioritisation and handoff	Queue coordination is locally negotiated rather than formalised	Task, Human	Bottleneck and fairness analyses miss practical prioritisation logic	Priority-change event with rationale
Cross-team coordination on patient readiness	Interdepartmental coordination spans systems and roles	Source, System, Task	Waiting-time drivers become invisible in case timeline analysis	Interdepartment status event

B.6 Textbook Figures Referenced in the Report

The figures below are included to support direct references in Part I and Part II. To separate them from assignment-produced figures, each caption explicitly preserves the textbook figure number.

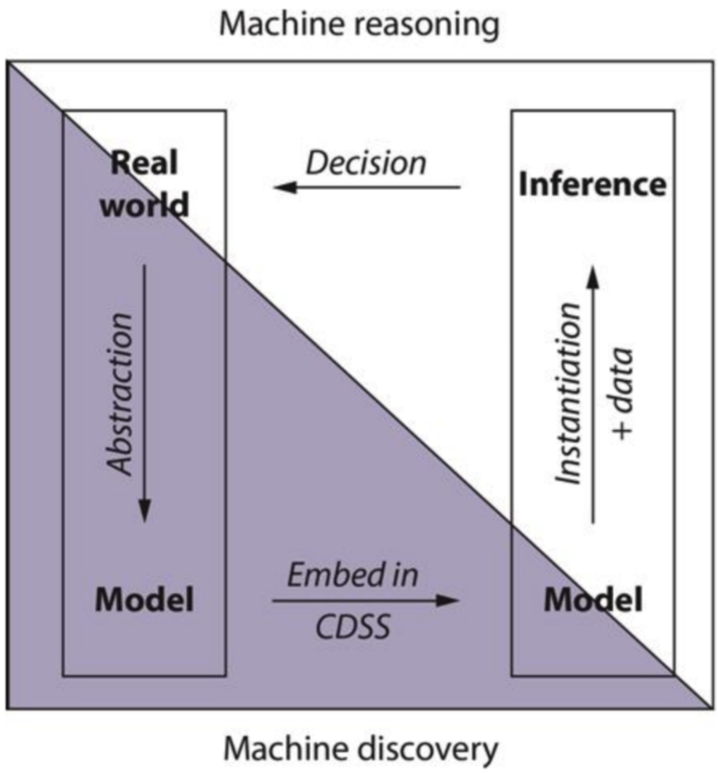

Figure 27.1 Computer reasoning depends first on acquiring models that record our understanding of how the world works and then applying those models in a decision support system to arrive at conclusions shaped by the data specific to a given decision task.

Figure 7: Textbook Figure 27.1 from *Guide to Health Informatics*: computer reasoning cycle from model construction through use

Source: Guide to Health Informatics (2026a). Reproduced from course compendium reading.

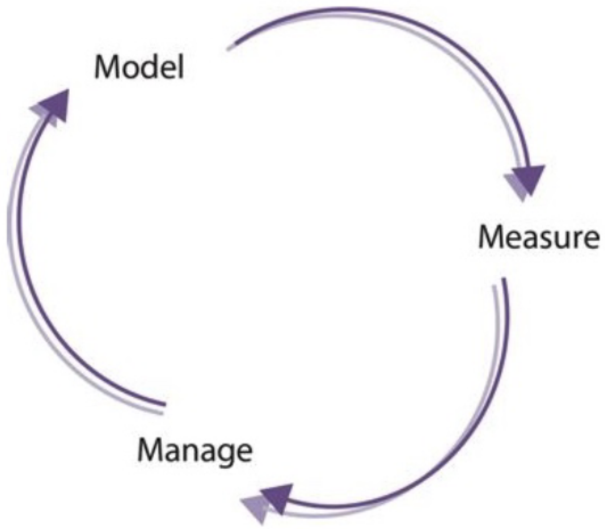

Figure 9.1 The model-measure-manage cycle. When the outcome of a management decision is fed back into another cycle, a feedback control loop is formed.

Figure 8: Textbook Figure 9.1 from *Guide to Health Informatics*: model-measure-manage cycle

Source: Guide to Health Informatics (2026b). Reproduced from course compendium reading.

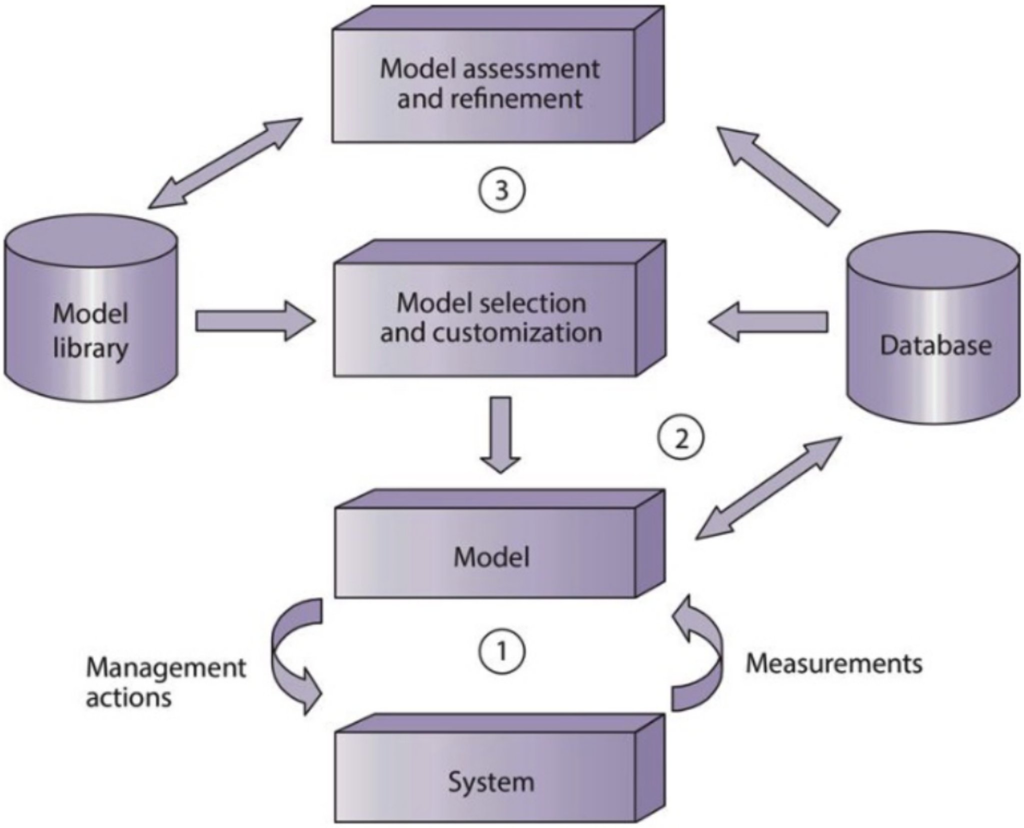

Figure 9.2 The three-loop model. Three separate information cycles interact within the health system. The first and second cycles select and apply models to the management of specific systems. The third information cycle tries to assess the effectiveness of these decisions and improve the models accordingly.

Figure 9: Textbook Figure 9.2 from *Guide to Health Informatics*: three-loop model

Source: Guide to Health Informatics (2026b). Reproduced from course compendium reading.

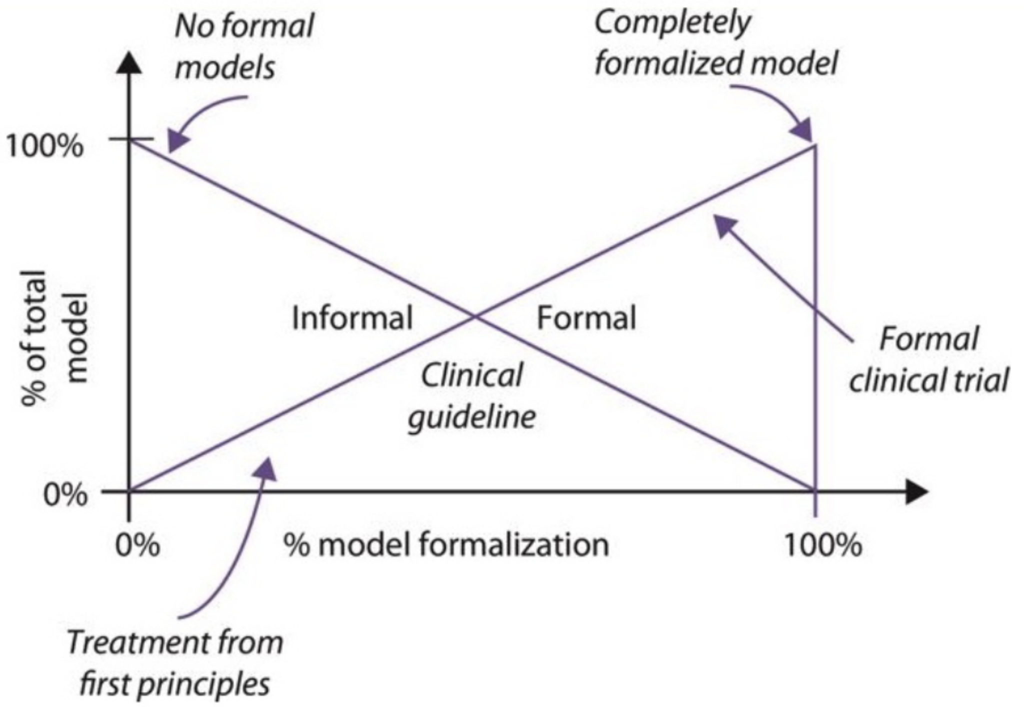

Figure 9.3 There is a continuum of possible formalization of information systems. Formal and informal models combine together to cover the total modelling needs of a given problem. The degree of formalization depends on the type of problem being addressed.

Figure 10: Textbook Figure 9.3 from *Guide to Health Informatics*: formalization continuum

Source: Guide to Health Informatics (2026b). Reproduced from course compendium reading.

References

Benson, T. and G. Grieve (2016). *Principles of Health Interoperability: SNOMED CT, HL7 and FHIR*. 3rd ed. Springer. ISBN: 978-3-319-30368-0.

Guide to Health Informatics (Course Compendium) (2026a). *Guide to Health Informatics: Selected Chapter 27*. PDF. Course reading PDF provided in NDAB20010U.

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Hildebrandt, T. T. (Feb. 2026). *Sundhedsdata og interoperabilitet: Process- and Protocol-aware Information Systems in Healthcare*. PDF. Course slides, NDAB20010U Health Data and Interoperability, University of Copenhagen.

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